



Fairfield University

School of Engineering & Computing

ASML

Project ISO: Tabletop Cleanroom

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Advisor: Dr. Andy Judge

Project Overview

Problem to be solved

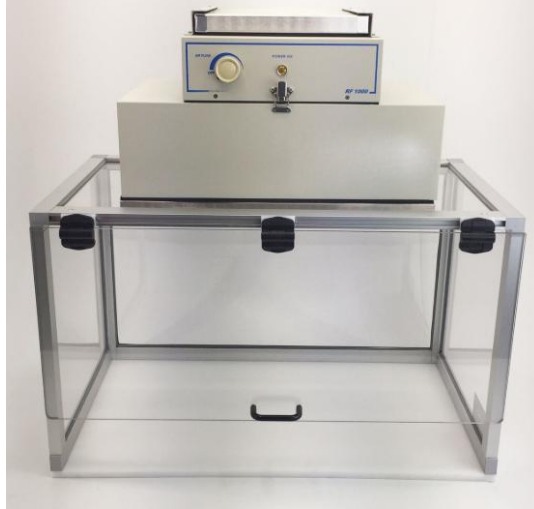
- Traditional cleanrooms - large and expensive
- Not easily accessible- limit research mobility

Solution

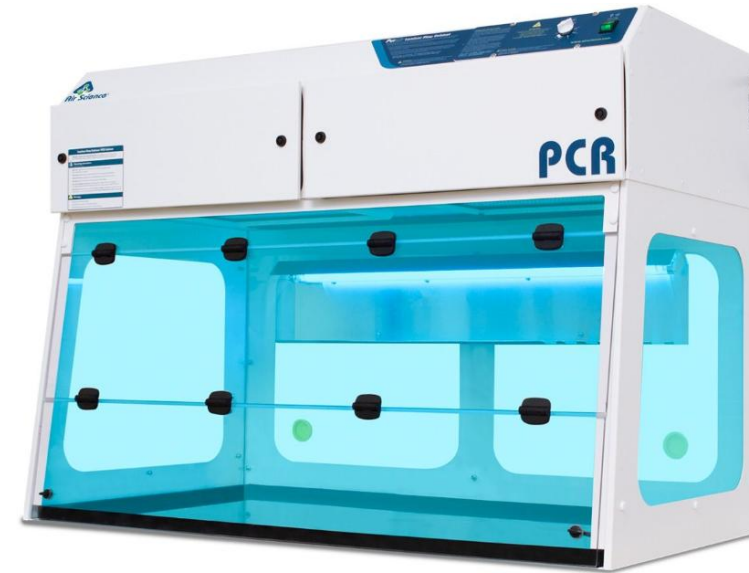
- Develop a portable, cost-effective tabletop cleanroom
- Provide Fairfield University with cleanroom research capabilities
- Establish a platform for future partnerships with organizations like ASML and NASA

Current on the market tabletop cleanrooms

- Compact, portable, and enclosed workspace designed to provide a particle-free environment- expensive and not sealed



Cleanroom World



Polymerase Chain Reaction (PCR) Workstation

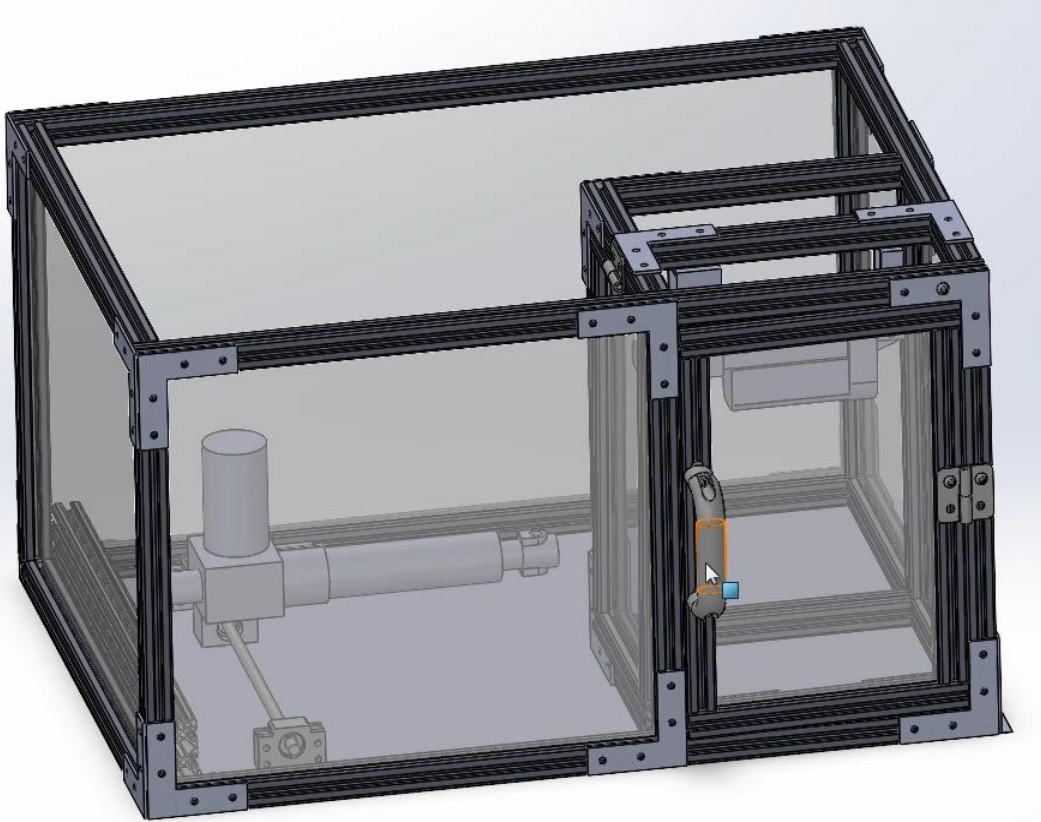


Objective

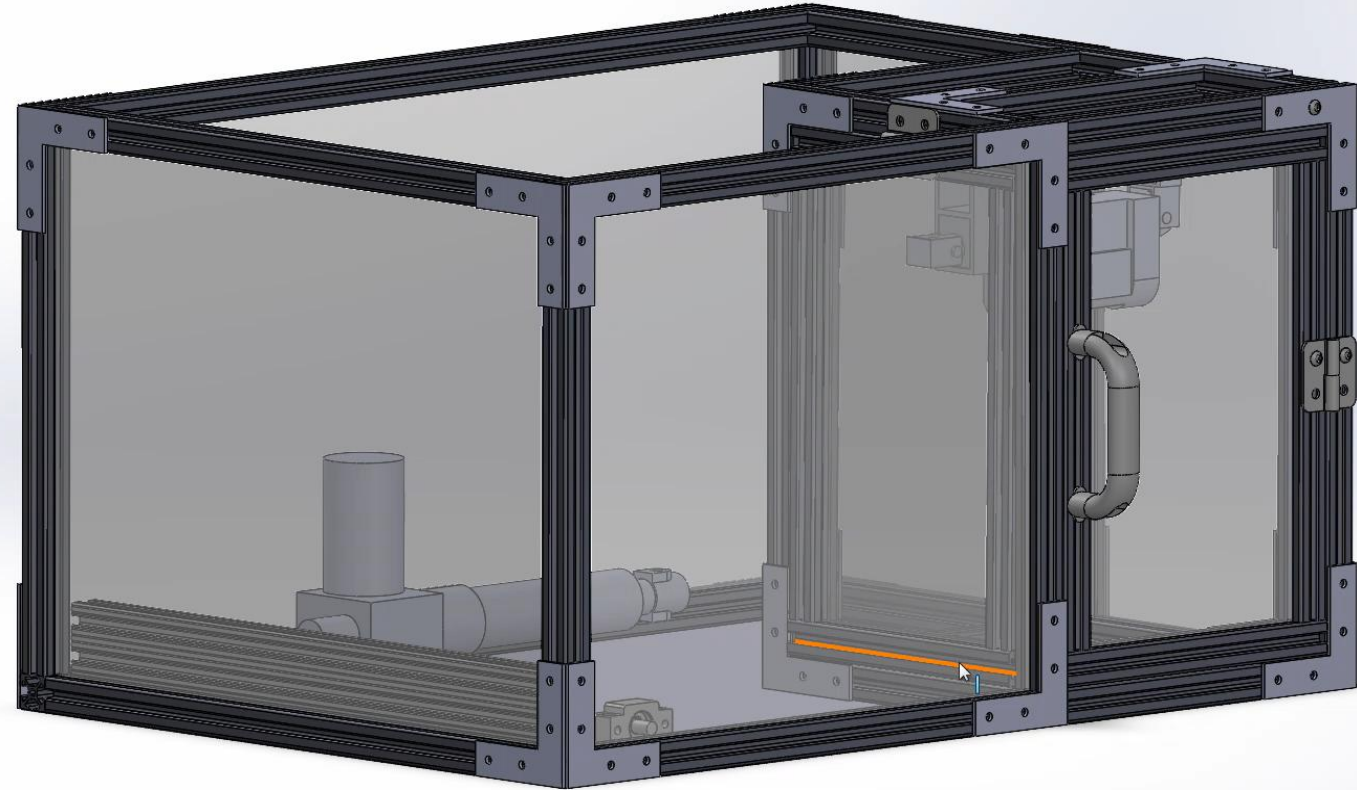
- Strengthens Fairfield's educational and research capabilities
- Achieve ISO 6 Standards
 - Within achievable budget
- Integrate mechatronics system into box to reduce particle contamination



Work Completed- Enclosure CAD



Manual Door

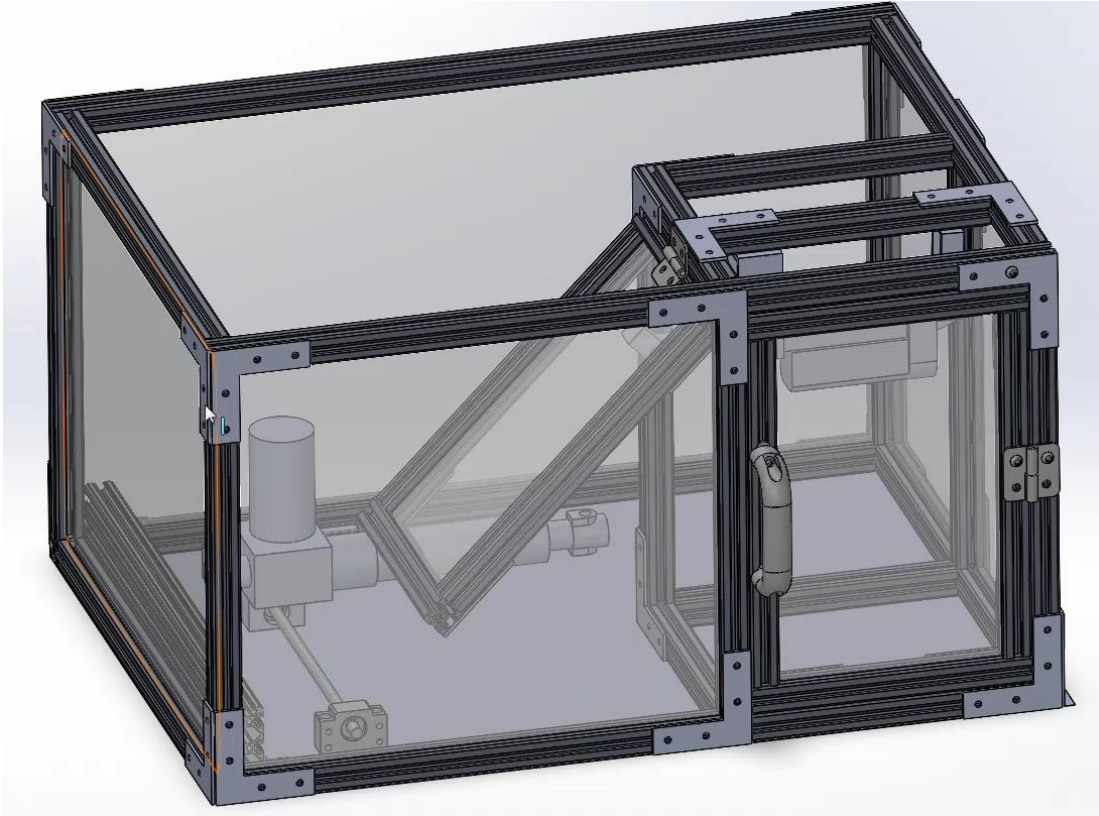


Automated Door

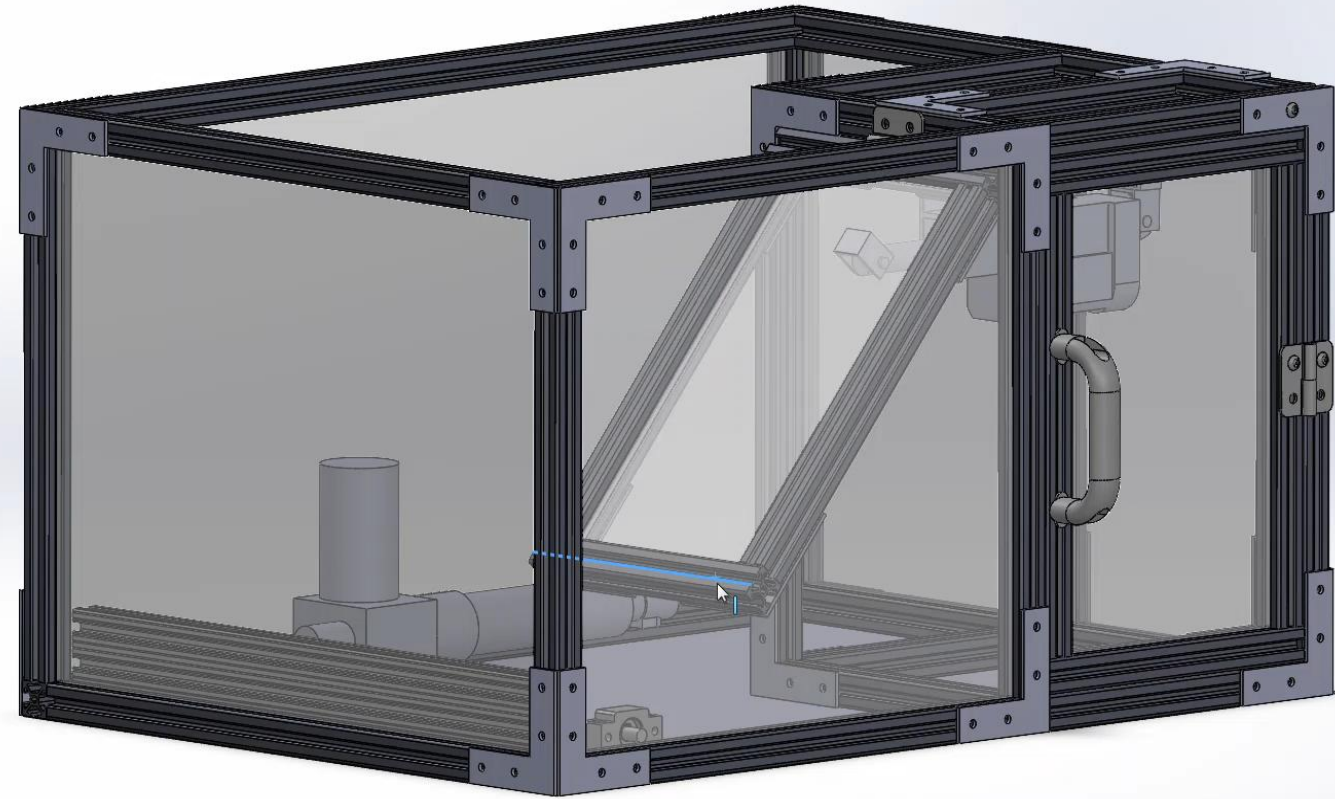




Work Completed- Enclosure CAD



Ball Screw and actuator



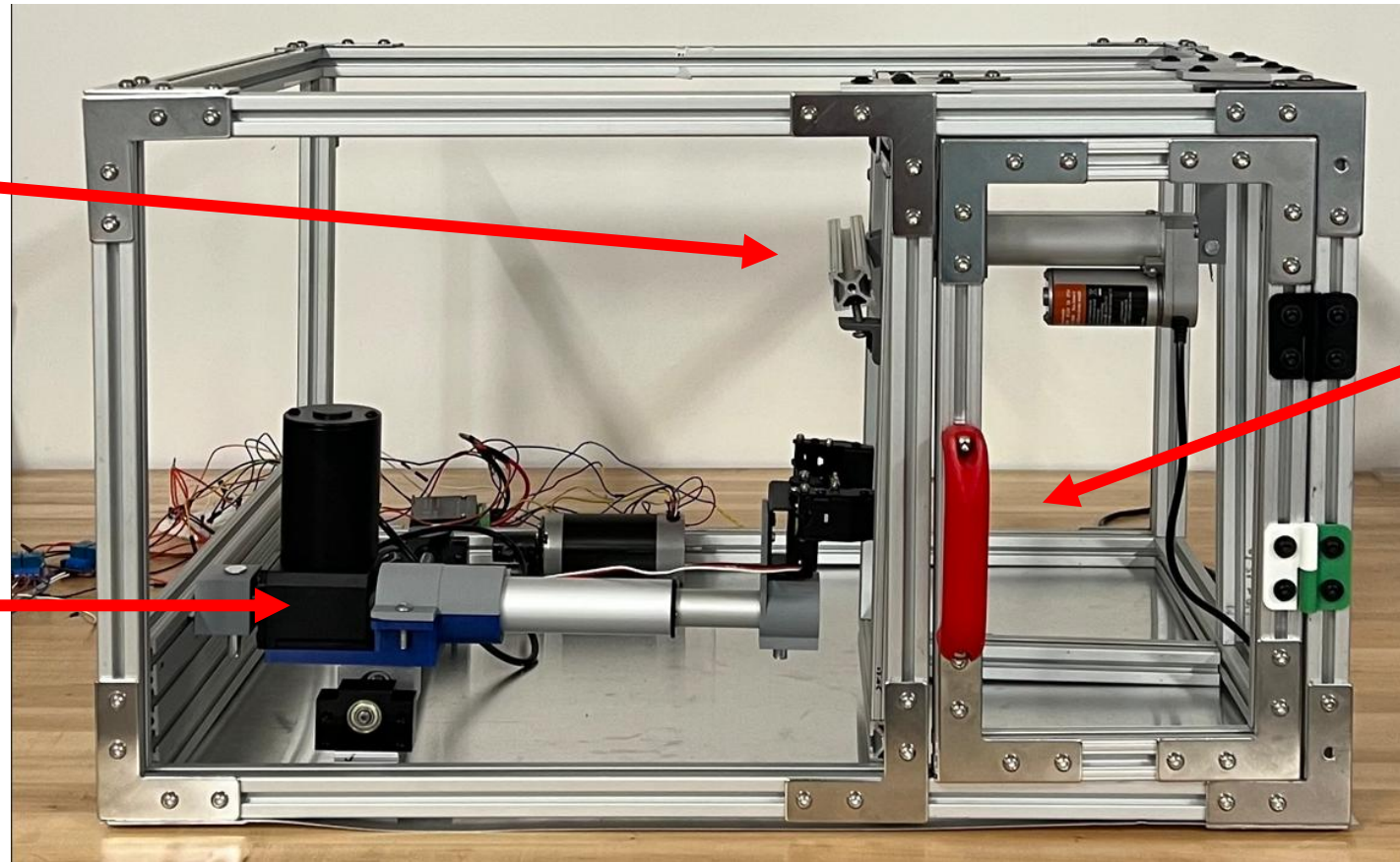
Automated Door



Work Completed- Enclosure early testing

Automated
Door

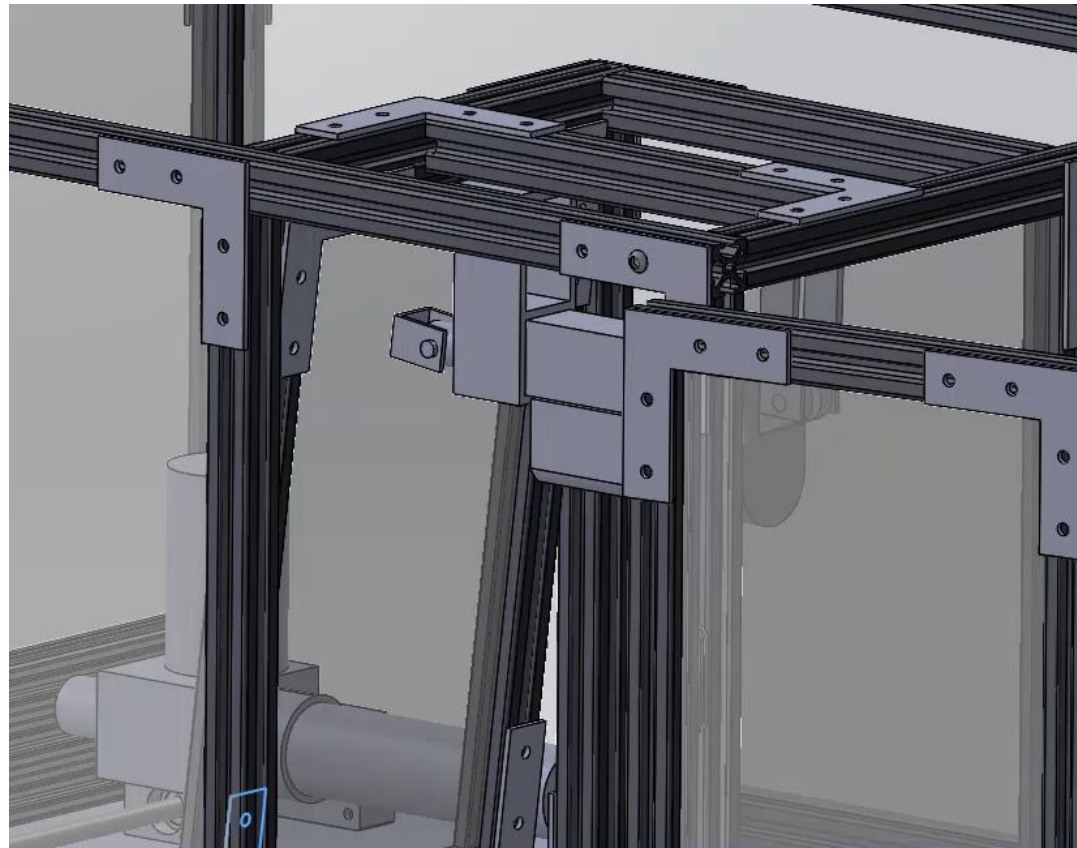
Manual
Door



Ball screw
and actuator

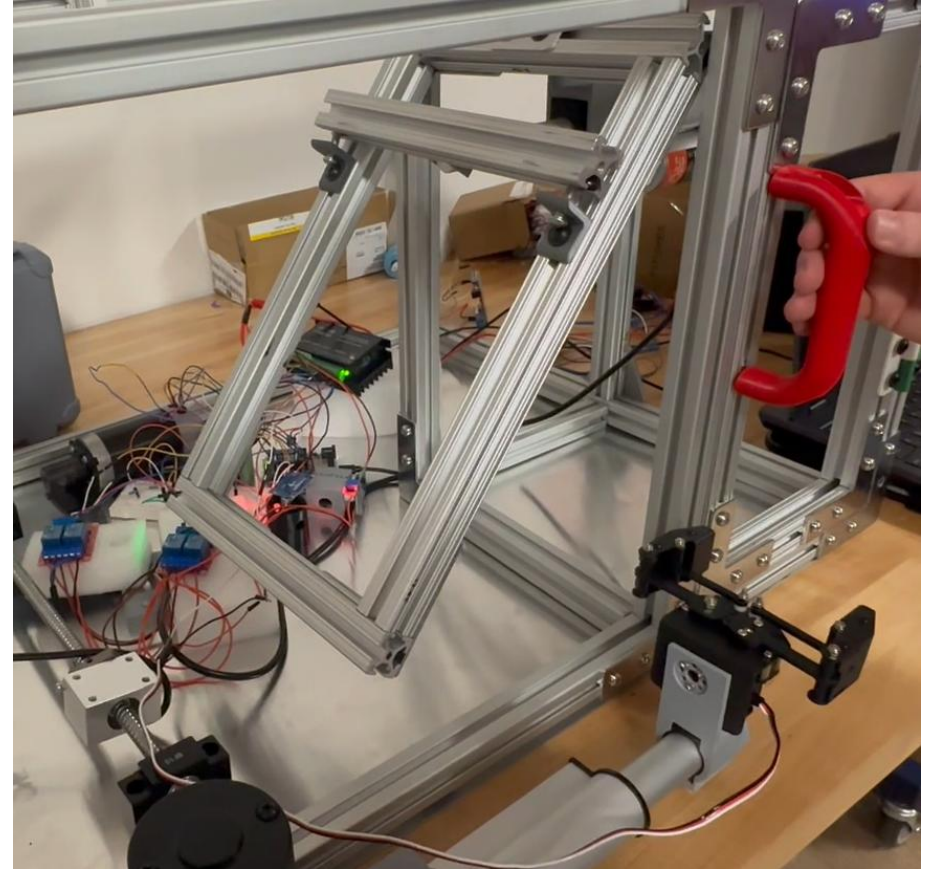
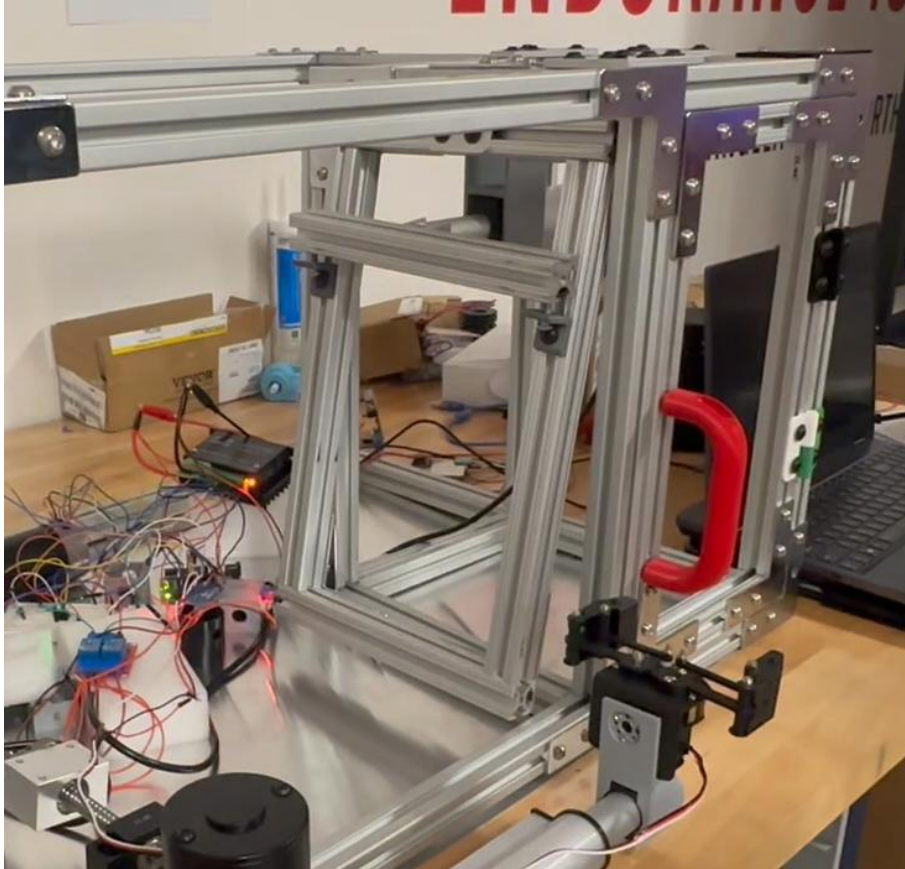


Work Completed- Mechatronics (actuator) CAD





Work Completed- Mechatronics (actuator) early testing



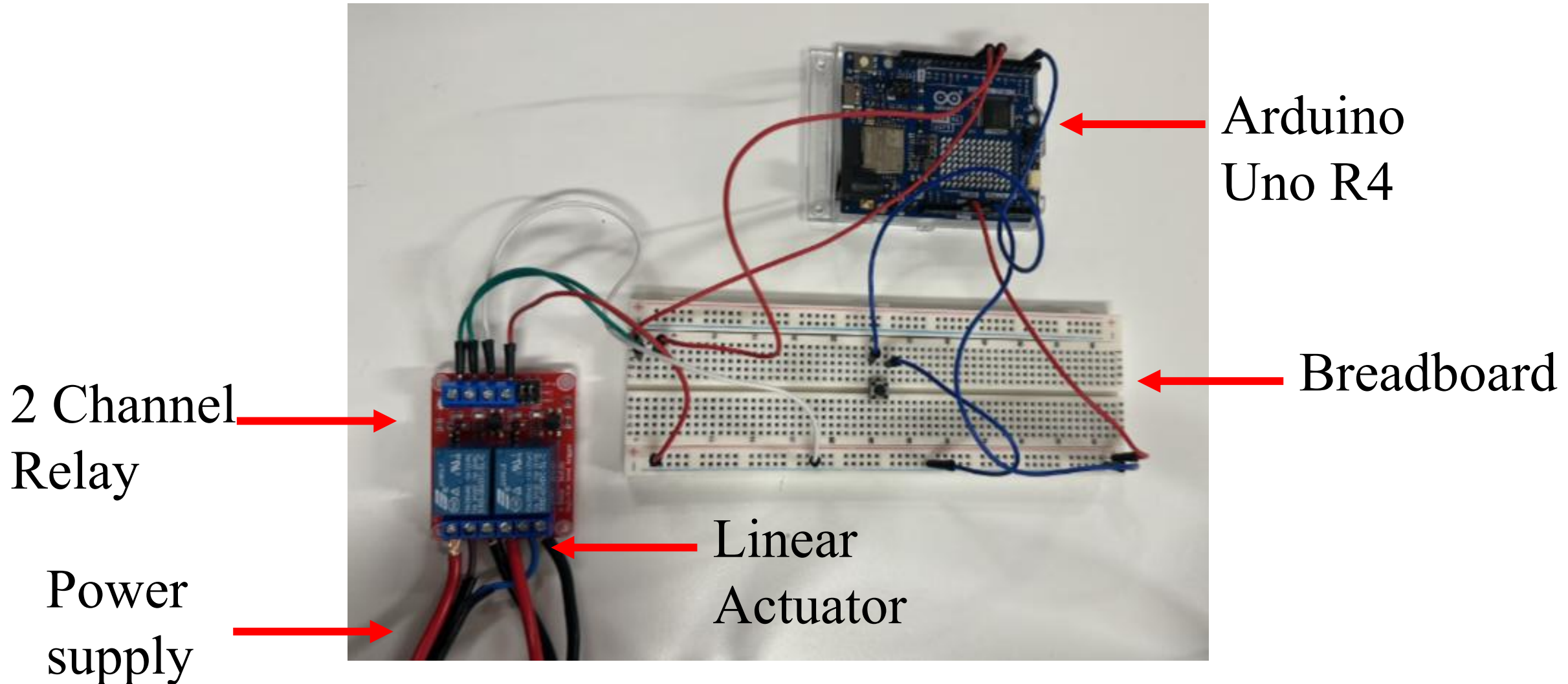
Automated Door Opening



Automated Door Closing

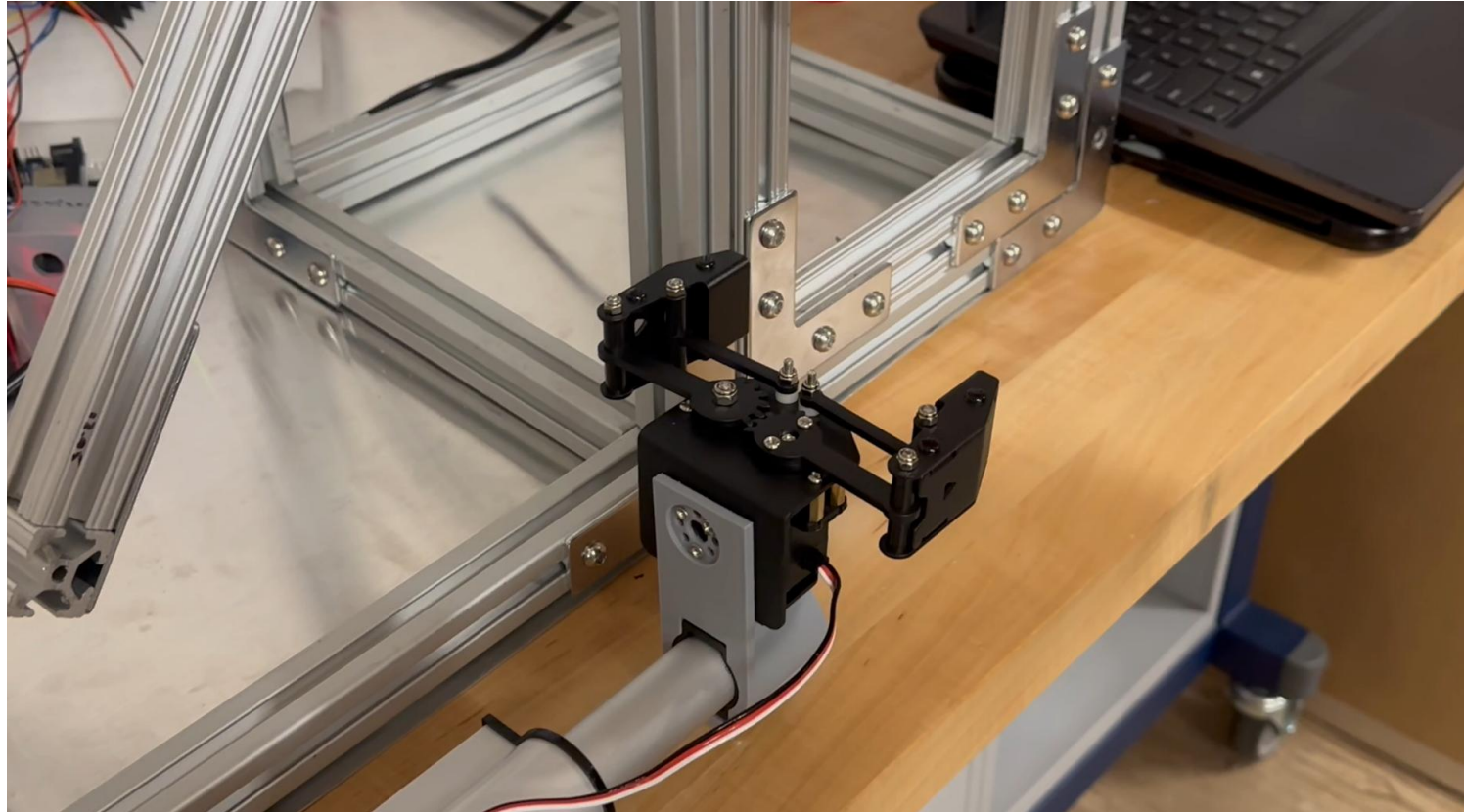


Work Completed- Mechatronics



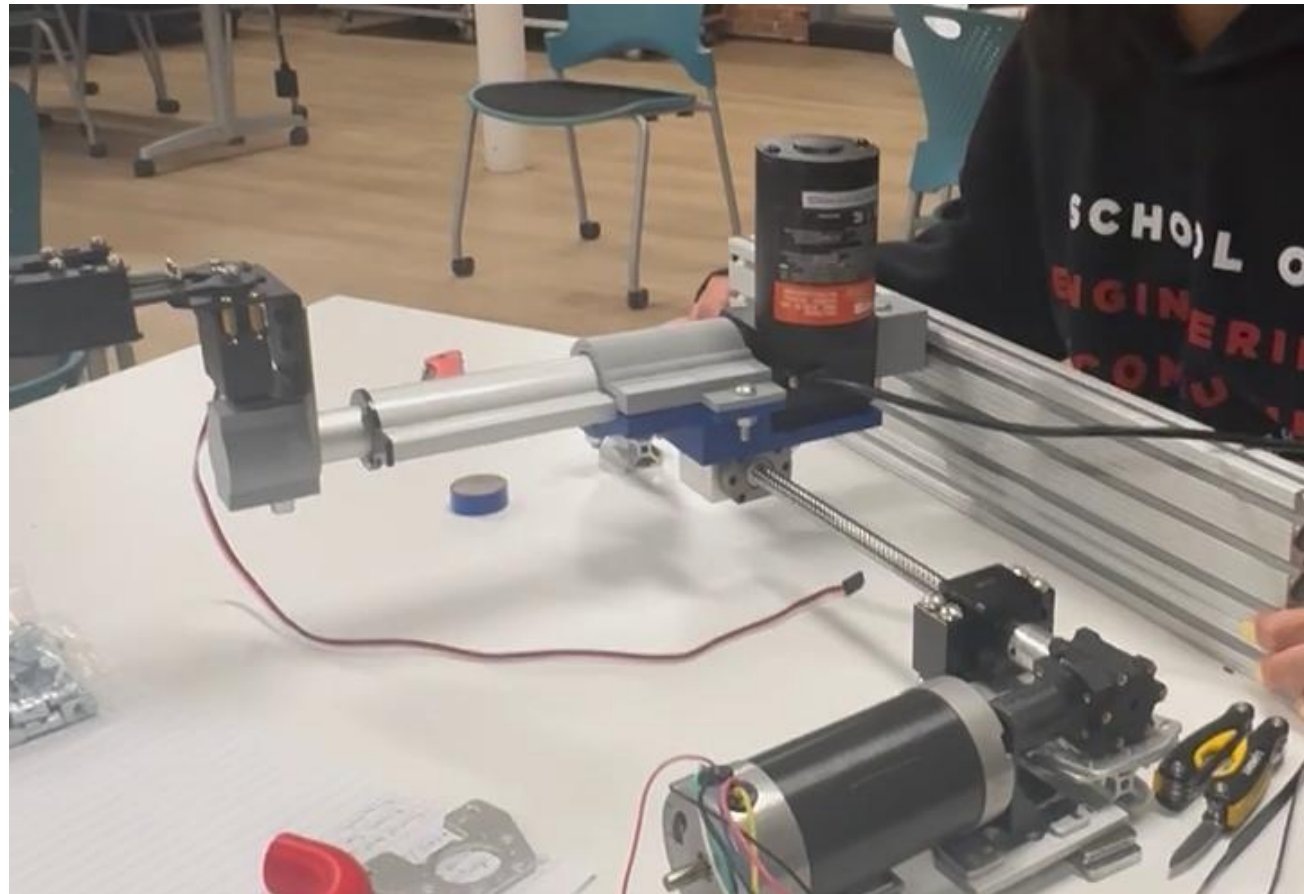


Work Completed- Early mechatronics (actuator + gripper) testing



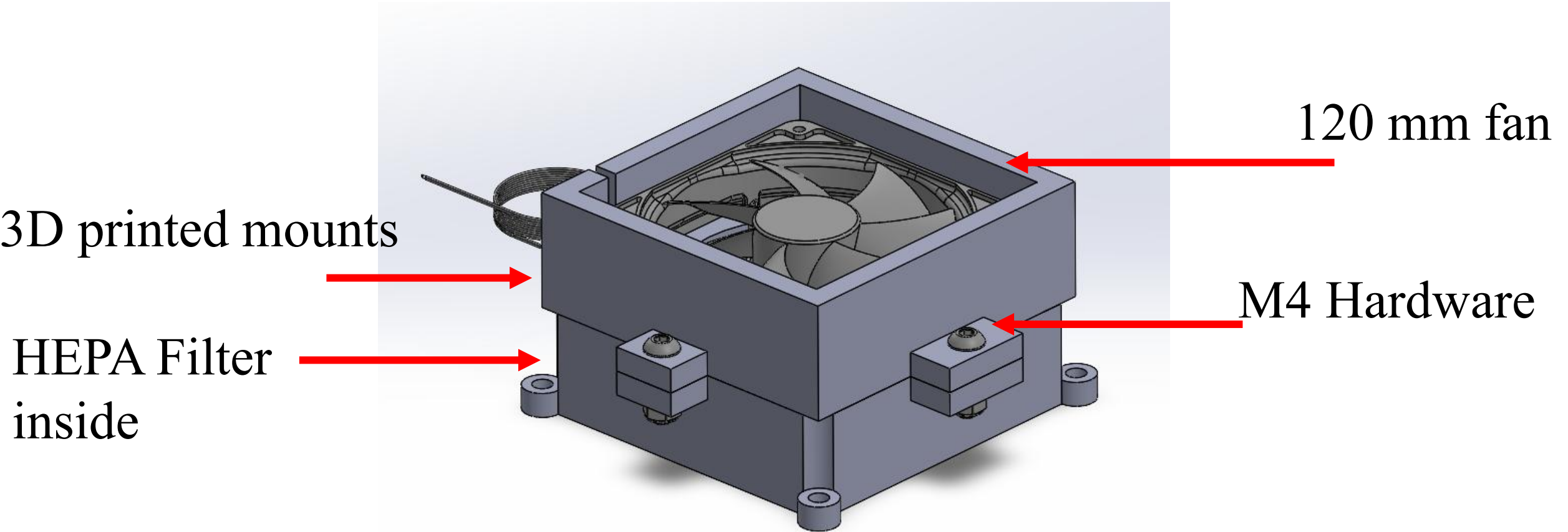


Work Completed- Early mechatronics (ball screw) testing





Work Completed- Airflow



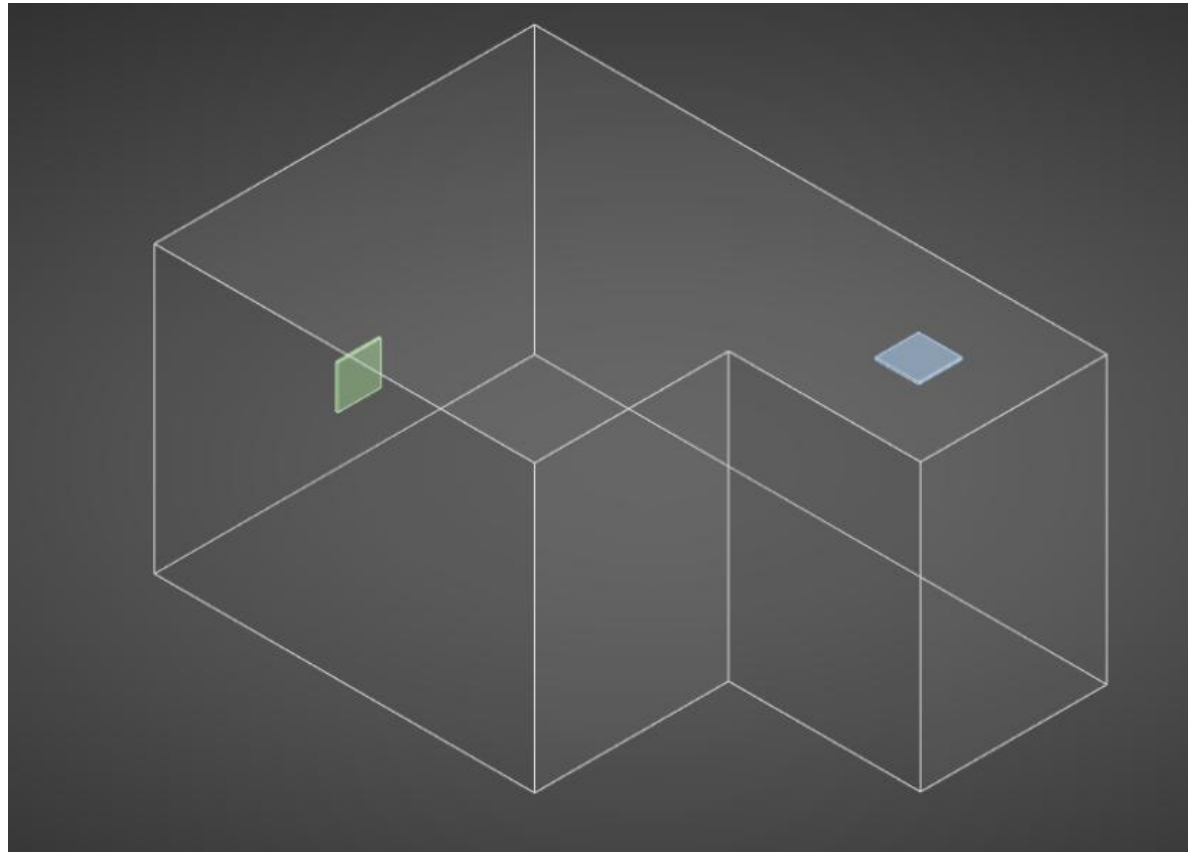


Work Completed- Airflow

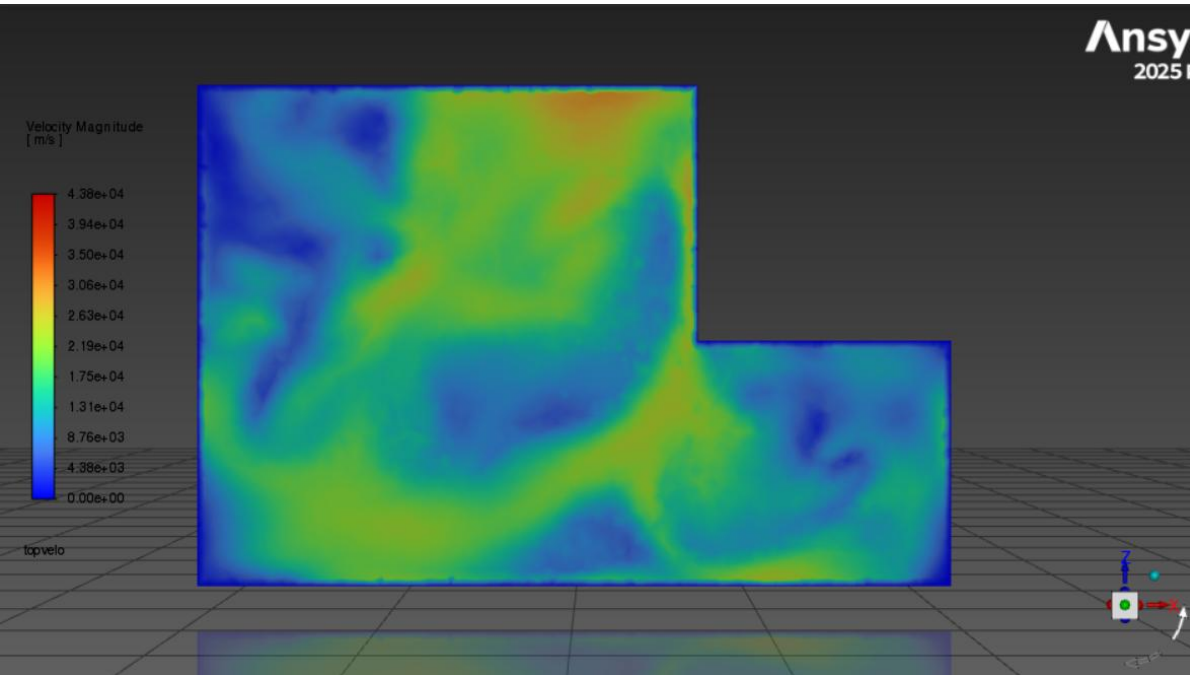




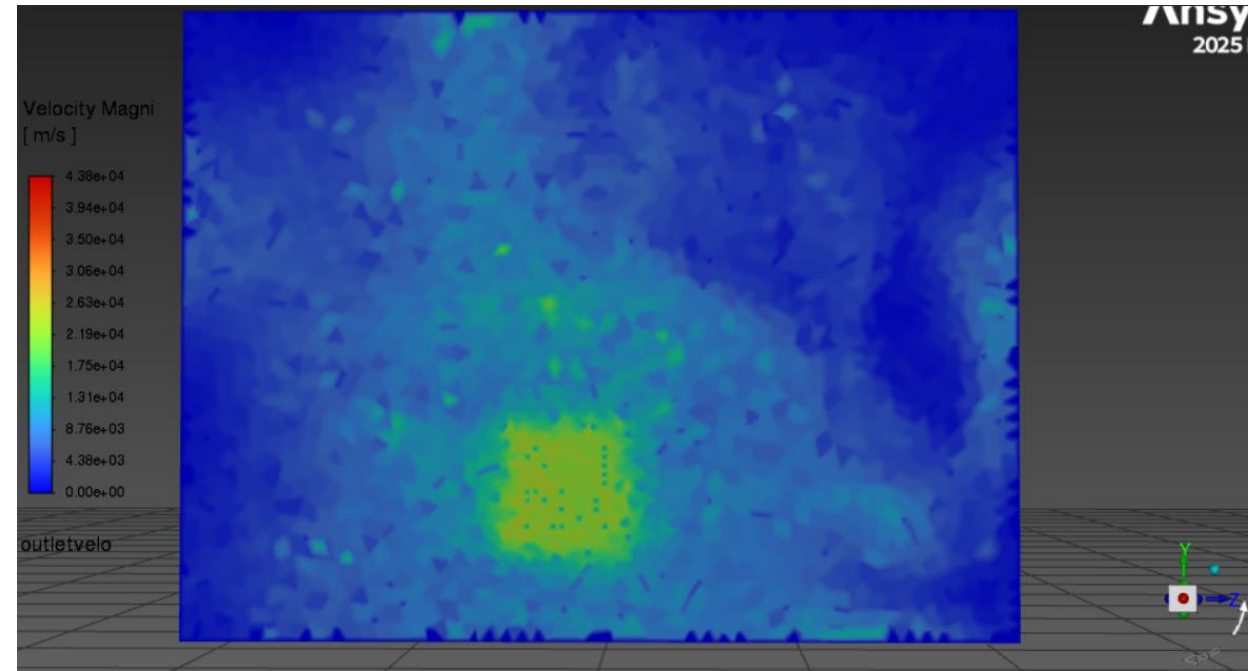
Work Completed - CFD



Work Completed - CFD



Top View (inlet)



Side View (exhaust)

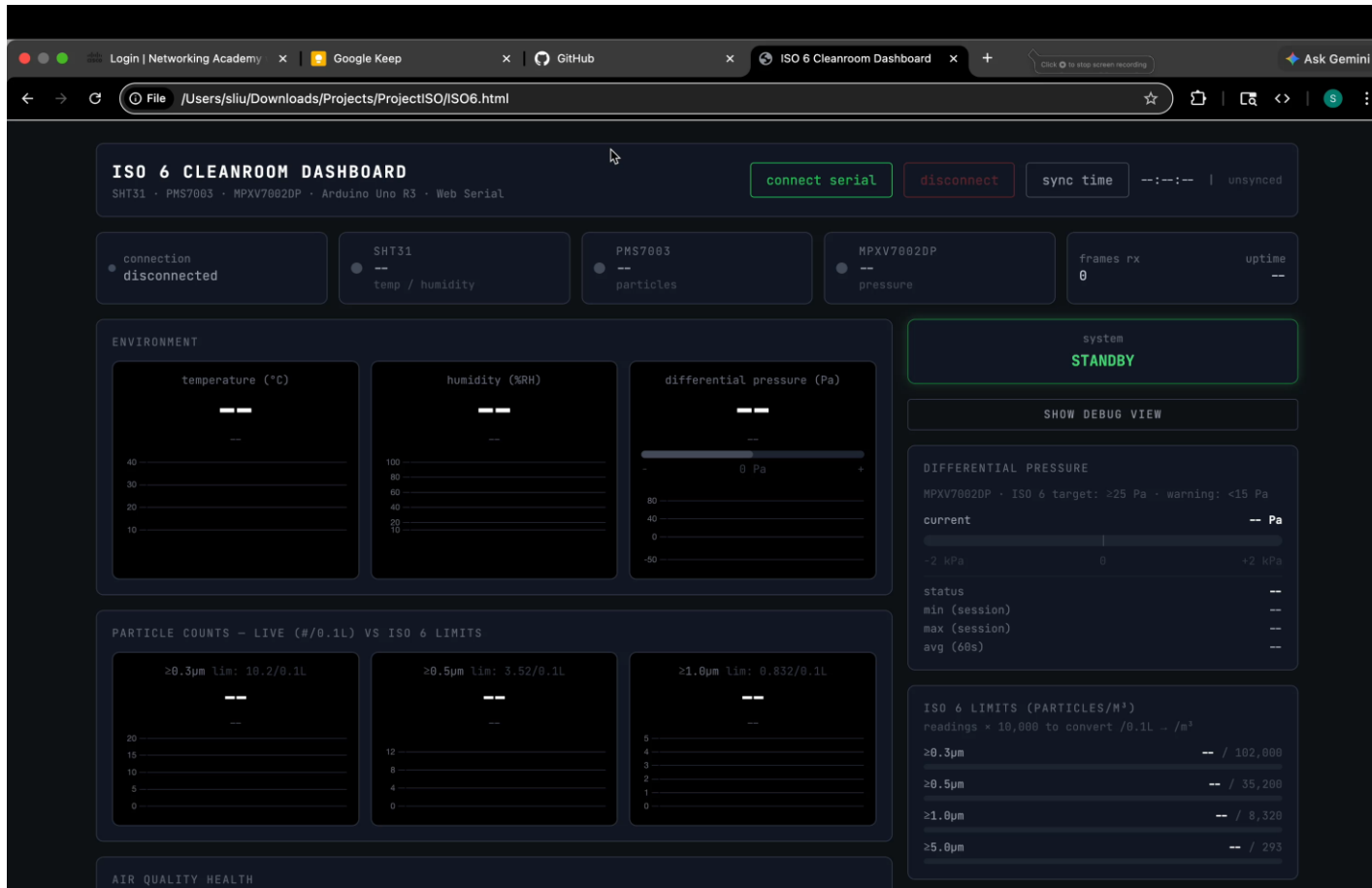
Low velocity



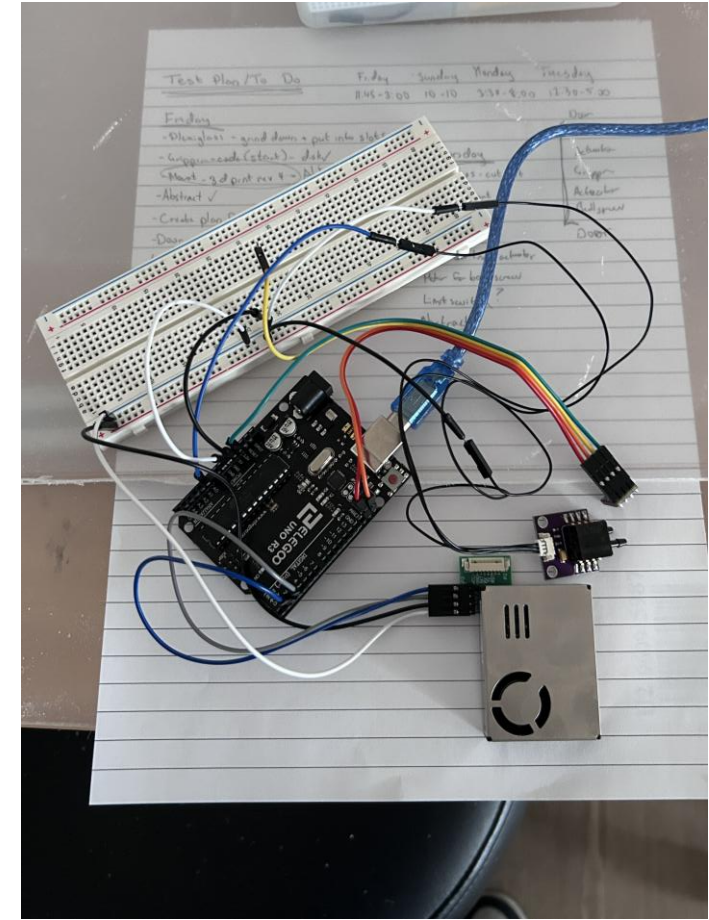
High velocity



Work Completed - Telemetry Dashboard



<https://youtu.be/jQzIZISJw5I>



<https://github.com/jooceboks/ProjectISO>



Challenges

Telemetry

- **Faulty Sensor**

- Defective unit found → replaced & recalibrated

- **Display Lag**

- Arduino too slow → moved logic to JS frontend

- **Power Overload**

- Sensors exceeded 5V rail capacity → rewired & used external power supply



Testing and results

Telemetry

- **Accuracy**
 - Dashboard correctly classifies air as *ISO Class 6*
- **Stability**
 - Real-time updates with no lag or freezing
- **End-to-End**
 - Sensor → Arduino → Web flow with zero data loss



Continued Work

- Enclosure

- Add plexiglass and sheet metal for walls and ceiling
- Mount fans and HEPA filters to sheet metal

-Ensuring proper airflow

- Proper fan speed adjustment
- Following CFD

-Dashboard

- Implement single click sequence automation for controls



Conclusion and Future Work

Milestone

- Successfully designed a custom tabletop cleanroom, with full testing and validation scheduled for completion this year

Next steps

- Proposed ASML project- cost effective particle inspection system



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Questions?

Thank you!